BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 033 · Passwords, Encryption and Penetration Testing

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. How does a firewall protect a network?

Q2. Which is an IPv6 address?
A. 2001:db8::7
B. 192.168.1.7
C. 02:11:22:33:44:55
D. school.example

Q3. What is a MAC address used for and how is it usually written?

Q4. Use decimal units. A file is 3 MB. Give its size in KB and bytes.

I Do · Read new knowledge

Key vocabulary

Authentication — Checking that someone is who they claim to be.

Strong password — Long and difficult-to-guess unique secret.

Two-factor authentication — Using two different forms of evidence to sign in.

Account lockout — Temporarily blocking access after repeated failed attempts.

Encryption — Transforming readable information into unreadable ciphertext.

Ciphertext — Encrypted form of data.

Penetration testing — Authorised testing for security weaknesses.

New learning

Strong passwords and authentication: Long, unique passwords make guessing harder. Multi-factor or two-step authentication adds a separate proof of identity.

Account lockout: Limiting repeated incorrect login attempts reduces opportunities for brute-force guessing.

Encryption: Encryption converts readable plaintext into ciphertext using an algorithm and key. Captured data is not readily understandable without the correct key.

Penetration testing: An authorised test looks for weaknesses so they can be fixed before attackers exploit them. It must be carried out with explicit permission and an agreed scope.

Worked example

A school wants to reduce the chance of attackers signing in with guessed passwords. Explain how strong passwords and account lockout help. [4 marks]

1. The risk is many attempted password guesses.

2. Long unique passwords increase the number of possibilities [1], making guessing the correct password harder [1].

3. A lockout policy blocks further attempts after repeated failures [1], reducing how many guesses an attacker can try [1].

4. Two protections, each explained with how the attacker is limited.

We Do · Practise with guidance

Guided 1

enter the member's account [1].

Q3. Strong passwords are difficult to guess [1], reducing unauthorised logins [1]. Encryption scrambles transmitted data [1], protecting contents if intercepted [1]. Authorised penetration testing identifies vulnerabilities [1], enabling patches/improvements [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It uses rules to allow or block network traffic.

Q2. A — 2001:db8::7

Q3. A MAC address identifies a network interface on a LAN. It is usually six pairs of hexadecimal digits.

Q4. (a) 3,000 KB and 3,000,000 bytes.

I Do / We Do pages · Guided practice

Q1. The login needs an additional independent authentication factor [1], such as a code on a separate device [1]. A stolen password alone is therefore insufficient [1], reducing unauthorised access [1].

Q2. The data is transformed from readable plaintext [1] into scrambled ciphertext [1]. If transmission is intercepted [1], attackers without the appropriate key cannot readily read the original information [1].

Q3. Authorised testers examine the system for vulnerabilities [1], using a defined scope [1]. They report weaknesses [1], allowing the organisation to correct them before malicious attackers exploit them [1].

You Do pages · Code First

```
failed = 4
if failed >= 3 then
    print("Lock account")
endif
```

Expected result: Lock account

You Do pages · Independent questions

Q1. Transforms readable plaintext [1] into unreadable ciphertext without the correct key [1].

Q2. Account lockout limits further login attempts after repeated incorrect passwords [1], making brute-force guessing less effective [1]. Two-factor authentication asks for a second independent proof [1], so guessing a password alone is insufficient to

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

A staff member accesses sensitive files away from school. Explain why two-factor authentication is useful. [4 marks]

Worked method: The login needs an additional independent authentication factor [1], such as a code on a separate device [1]. A stolen password alone is therefore insufficient [1], reducing unauthorised access [1].

Guided 2

A website sends account information across the Internet. Explain why encryption is important. [4 marks]

Worked method: The data is transformed from readable plaintext [1] into scrambled ciphertext [1]. If transmission is intercepted [1], attackers without the appropriate key cannot readily read the original information [1].

Guided 3

A college hires authorised testers before launching an online system. Explain the purpose and two benefits of penetration testing. [4 marks]

Worked method: Authorised testers examine the system for vulnerabilities [1], using a defined scope [1]. They report weaknesses [1], allowing the organisation to correct them before malicious attackers exploit them [1].

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict the account response.

```
failed = 4
if failed >= 3 then
    print("Lock account")
endif
```

predict

Question 1

State what encryption does to data. [2 marks] Source: Adapted from your Paper 1 revision booklet, 1.4.2.

Question 2

An online sports club membership portal is targeted by repeated password guesses. Explain how account lockout and two-factor authentication can help protect member accounts. [4 marks] Source: Original J277-style question; not an OCR past-paper original.

Question 3

Explain how strong passwords, encryption and penetration testing improve network security. [6 marks] Source: Original J277-style question; not an OCR past-paper original.

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Exit Ticket · Answer before checking

Penetration testing should be conducted only with permission from the system owner.